

Quick Check

Lesson 3: Following the Fluid: How the Nephron Filters Blood and Makes Urine

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.3: Excretion and Homeostasis in Animals | Lesson 3

Blood flows into the glomerulus in a mammalian kidney. In which part of the nephron does ultrafiltration take place?

- A** Proximal convoluted tubule, where glucose is taken back
- B** Collecting duct, where ADH controls water movement
- C** Loop of Henle, where the medulla is made concentrated
- D** Glomerulus and Bowman's capsule, where fluid is pushed out of the blood

Blood cells and large proteins are found in the blood entering the glomerulus but not in the glomerular filtrate. What is the best explanation?

- A** They are too large to pass through the filtering membrane under pressure
- B** They are secreted into the distal tubule and leave in urine
- C** They are broken down by ADH inside Bowman's capsule
- D** They are filtered out first but are reabsorbed in the proximal tubule

A healthy person eats ugali and their blood glucose rises. Glucose is filtered into the nephron, yet none is found in the urine. What happens to the glucose?

- A** It is carried to the bladder but is too dilute to detect
- B** It is all reabsorbed into the blood in the proximal tubule using carrier proteins and energy
- C** It is changed into urea in the loop of Henle
- D** It is destroyed in Bowman's capsule before reaching the tubule

In the 'Follow the Fluid' table, urea is present in the glomerular filtrate and is much more concentrated in urine. Why is urea more concentrated in urine than in the filtrate?

- A Water is reabsorbed along the nephron, but most urea stays in the tubule
- B Urea is reabsorbed into the blood faster than water
- C ADH changes the water into urea
- D Extra urea is made in the collecting duct

On a hot day a goat loses water through sweating and has not drunk much. Its brain releases more ADH. What does this cause in the kidney?

- A** The proximal tubule stops reabsorbing glucose
- B** The collecting duct reabsorbs less water, so a large volume of dilute urine is produced
- C** The glomerulus filters more proteins into Bowman's capsule
- D** The collecting duct reabsorbs more water, so a small volume of concentrated urine is produced

A freshwater fish takes in a lot of water by osmosis through its body surface. Which description of its kidney activity fits its dilute urine?

- A** It excretes the excess water as a large volume of dilute urine
- B** It stops filtering blood so that no water enters the nephron
- C** It makes urine only when it is out of the water
- D** It reabsorbs almost all the water, so the urine is concentrated

Which statement correctly contrasts ultrafiltration with selective reabsorption in the nephron?

- A** Ultrafiltration is pressure-driven and depends on size, while selective reabsorption returns useful substances using carrier proteins and energy
- B** Ultrafiltration uses carrier proteins and energy, while selective reabsorption is driven by blood pressure
- C** Ultrafiltration happens in the collecting duct, while selective reabsorption happens in the glomerulus
- D** Both processes are driven only by blood pressure and work in the same part of the nephron

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).